# Nathan Liu

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## Education

**New York University** — New York, NY

*Bachelor of Arts in Computer Science and Data Science | Dean's List* — *Sep. 2024 – May 2028*

Relevant Coursework: Probability and Statistics, Principles of Data Science I & II, Data Management and Analysis, Fundamentals of Machine Learning, Linear Algebra, Causal Inference (in progress)

## Experience

**Data Science Intern** — Nov. 2025 – Present

*NYU Entrepreneurial Institute* — *New York, NY*

- Built an LLM-assisted pipeline (Perplexity, Google Search APIs) classifying 2.6k+ startup records by NYU school for segment reporting
- Shipped 5 features to a BigQuery-backed React analytics dashboard tracking 2.6k+ annual coaching sessions and 5k+ visits, including drill-down time-series charts and year-over-year comparison
- Audited 31 production data pipelines against documentation, tracing 24 discrepancies to root causes and flagging 7 untracked scripts writing to production data
- Leading CRM deduplication of ˜14k entries, building a 50-row gold set to QA duplicate-detection signals

**Full-Stack AI Developer** — June 2025 – Aug. 2025

*Amazon China*

- Trained a scikit-learn regression model on 30-day rolling prices for short-term forecasting in an AI trading assistant (Flask, React)
- Built a signal engine combining TextBlob sentiment on 40 headlines per refresh with 24h price change to classify assets as buy, sell, or hold
- Validated sentiment classification against manually labeled headlines in Jupyter and wrote 26 tests (pytest, Jest)

**Outreach Coordinator** — Feb. 2026 – Present

*NYU Data Science Club* — *New York, NY*

- Co-organized NYU's 2026 Datathon for 50+ participants, leading sponsor outreach to companies including Datadog and TikTok

## Projects

**HotTakes** | *2nd Place, NYU Stern Sustainability Hackathon* — Oct. 2026

- Trained a gradient-boosted heat demand forecast (scikit-learn) on synthetic hourly data calibrated to Con Ed filings, reaching 8.5% MAPE vs 17.5% for a naive baseline, and presented the model in the pitch
- Validated on real metered steam from 7 buildings, cutting WAPE to 18% vs 24% and 39% for two baselines
- Compared 5 outage reserve policies in an 8,760-hour simulation, finding a reserve does not pay since storage is usually full
- Deployed a Streamlit dashboard (5 tabs, 20 inputs) with a Claude assistant that calls 4 read-only model tools

**Ad Click-Through Rate Prediction (Fundamentals of ML)** | *Python, pandas, scikit-learn* — Summer 2026

- Predicted clicks on 1M Avazu ad impressions, reaching 0.723 ROC-AUC and 0.411 log loss vs a 0.456 base-rate baseline
- Engineered 105 leakage-free features, adding time features after EDA showed click rates varied by hour and device
- Showed why accuracy misleads on imbalanced data: a never-click model scores 83% accuracy but 0.50 ROC-AUC

**MarginGuard** | *Python, DuckDB, Claude API* — Apr. 2026

- Built a Python/SQL pipeline in DuckDB joining 1.2M+ labor logs and 90K+ billing records into project-level signals
- Defined margin metrics and erosion thresholds an AI agent uses to rank the 3 highest-exposure of 405 projects

## Technical Skills

**Languages**: Python, SQL, R, Java, C/C++, JavaScript
**Data & ML**: pandas, NumPy, scikit-learn, XGBoost, Matplotlib, Seaborn, Jupyter, Streamlit
**Data & Cloud**: BigQuery, DuckDB, PostgreSQL, Google Cloud Run, Git/GitHub
**Methods**: Regression and forecasting, classification, clustering, statistical inference, causal inference, root-cause analysis